

Forward Reasoning in Isabelle (FW)

Lecture Notes Transcription

Course: Interactive Theorem Proving (7CCSACTP)

1. Context & Overview

→ **Last Video Revision:**

- **Basic Functional Programming:** Currying, Lambda expressions, Polymorphic types.
- **Isabelle Interface:** Theory files, terms (numbers, lists, sets, products/tuples, functions) and type annotations.
- **Logical Specifications:** Connectives and theorem structure.

→ **Performing Proofs in Isabelle:**

- **Logical Reasoning:** Forward reasoning (FW) & Backward reasoning (BW).
- **Proofs by Substitution.**
- **Mathematical Induction** on natural numbers (**nat**).

2. Forward Reasoning Mechanics: Example Proof

Goal Statement: $P \rightarrow Q, \quad Q \rightarrow R, \quad \neg R \vdash \neg P$

Step-by-Step Derivation:

1. $\neg R$ (*Assumption 1*)
2. $Q \rightarrow R$ (*Assumption 2*)
3. $Q \rightarrow \neg R$ (*from 1, using $\rightarrow I$ rule*)
4. $\neg Q$ (*from 2, 3, using $\neg I$ rule*)
5. $P \rightarrow \neg Q$ (*from 4, using $\rightarrow I$ rule*)
6. $\neg P$ (*from assumption $P \rightarrow Q$ and 5, using $\neg I$*)
✓

Inference Rules & Isabelle Syntax:

- Implication Introduction ($\rightarrow I$):

$$\frac{X}{Y \Rightarrow X}$$

- Negation Introduction ($\neg I$):

$$\frac{X \rightarrow Y \quad X \rightarrow \neg Y}{\neg X}$$

- **Isabelle Forward Attribute (OF):**

$$f_2 \text{ [OF } f_1]$$

Instantiates premise of rule f_2 with known fact f_1 (e.g. `impI[OF fact1]`).

3. Isabelle Forward Reasoning Notes

- **Using Facts:** To obtain and utilize a fact P in Isar:

`have P    using P`

- **Schematic Variables:** In Isabelle, schematic variables written with a question mark (e.g. `?x`) represent universally quantified variables ($\forall x$).